

Cross Entropy Loss.

Loss function:

- loss functions are used to optimize the model during training.
- objective → minimize the loss function.
- lower the loss the better the model.
- Cross-Entropy loss is a most important cost function.
- used to optimize classification models.

Loss function:

☐ Cross Entropy.

☐ Hinge

☐ Huber

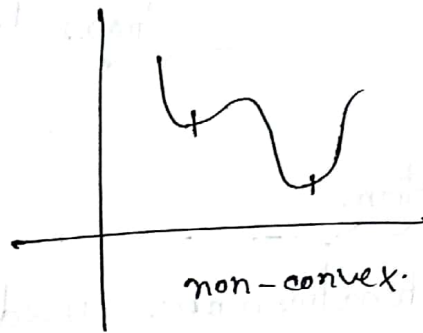
☐ Kull back - Leibler.

☐ RMSE

☐ MAE.

☐ MSE.

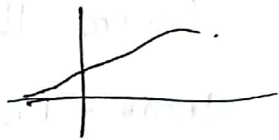
Cross Entropy loss:



$$\text{loss} = |y - \hat{y}| \rightarrow \text{linear.}$$

$$\text{Euclidean distance} = |y - \hat{y}| \rightarrow 2|y - \hat{y}| = 2|y - \hat{y}|.$$

activation $\rightarrow$ non-linear. ଏହା ଫାଟର sigmoid graph



So, $\text{loss} \propto \text{linear}$ ଓ; activation $\propto$ non-linear. ହେତୁ ଏହାମାନଙ୍କ ମିଶ୍ରଣ ହେଉଛି non-convex ଫାଟର (ଏହା ସତ୍ୟ)। ଏହା local minima ପ୍ରାପ୍ତ ହେବା ସମ୍ଭାବନା ରଖେ।

ଏହି ଆମାନ୍ତର loss function ଏକ non-linear ଫାଟର ଫାଟର।
ଏହା activation function $\propto$ non-linear। So,
non-linear output ଓ; measurements non-linear ଫାଟର convex.
ପ୍ରାପ୍ତ ହେବ।

ଏହି non-linear measurement $\rightarrow$ ଏହା cross Entropy loss / logarithmic loss.

Cost: ଏହା ଆମର input ଓ; loss ଓ; average $\rightarrow$ ଏହା
cost.

Entropy: तंत्रज्ञान

probability

$$\begin{array}{|c|} \hline \square \\ \hline \end{array} \rightarrow 1 \text{ (I)} \rightarrow \frac{1}{2}$$

$$\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \rightarrow 2 \text{ (I)} \rightarrow \frac{1}{4}$$

$$\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \rightarrow 3 \text{ (I)} \rightarrow \frac{1}{8} \rightarrow \frac{1}{2^3} \rightarrow \frac{1}{2^I}$$

Information theory.

Information: What is the minimum number of Yes/No Qs. you have to ask to reach a outcome.

$$2^I = \text{no of box} \quad \text{--- (I)}$$

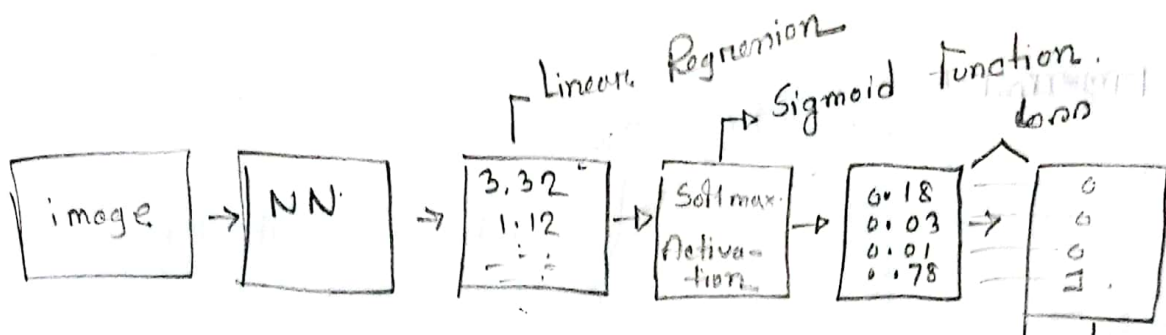
$$\text{probability} = \frac{1}{\text{no of box}} \quad \text{--- (II)}$$

(I) and (II) together,

$$2^I = \frac{1}{P}$$

$$\Rightarrow \log_2 (2^I) = \log_2 \left(\frac{1}{P} \right) \rightarrow \frac{1}{P} = P^{-1}$$

$$\Rightarrow I = -\log_2 P \rightarrow \text{Information Theory.}$$



Logits: it is non-normalized.

→ output before it goes to activation function.
going to a softmax activation function is called logits / non-normalized output.

→ input of softmax.

Softmax - Activation function:

→ multilevel probability for softmax.

→ Binary / Sigmoid (for result after 1 bit) → sigmoid.

softmax for probability to normalized for.

for sigmoid for 2 probability or any other.

Binary / multiple class for any → softmax.

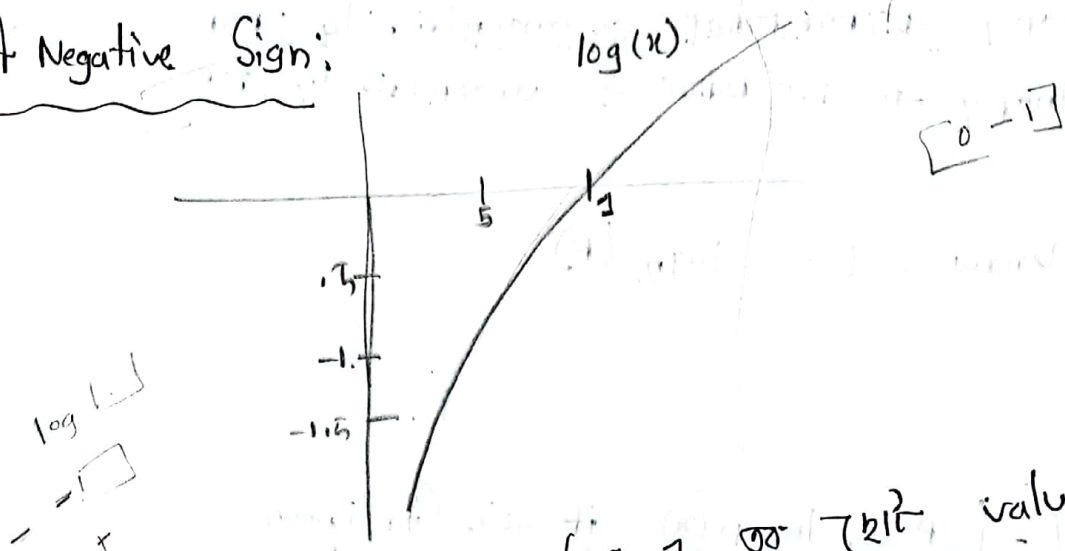
Binary class for any → sigmoid. → 1 bit output.

Logits: the vector of raw (non-normalized) predictions that a classification model generates, which is ordinarily then passed to a normalization function. If the model is solving a multi-class classification problem, logits typically become an input to the softmax function. The softmax function then generates a vector of (normalized) probabilities with one value for each possible case.

Entropy: Entropy provides a measure of the average amount of information needed to represent an event drawn from a probability distribution for a random variable.

For random variable (ଅନନ୍ତ ସମ୍ଭାବ୍ୟତା) individual probability distribution
 ଏକ ସମ୍ଭାବ୍ୟତା ବଣ୍ଟନର entropy (ଅବସ୍ଥା) ଗଣନା କରିବାକୁ ହେବ।

Reason of Negative Sign:



ଏହି logarithmic graph (ଯାହା ଧନାତ୍ମକ 1 ଠାରୁ ଛୋଟ value ଉପରେ) ଏହା ଏହା value negative.

ଆମେ $\log(p)$ measure କରି ଏହା ଧନାତ୍ମକ 1 ଠାରୁ ଛୋଟ ହେବା ସ୍ୱାଭାବିକ ନାହିଁ।

ତେଣୁ Ans ଏହା (-) ନା ଆମେ ଏହି equation ଉପରେ ଆମେ (-) ଲେଖିବା।

Q10 container 2 ଠାରୁ entropy ଗଣନା କରିବା।

ଆମେ ଗଣନା କରିବା। Δ ସମ୍ଭାବ୍ୟତା probability = $\frac{14}{30}$

$$0 \quad n \quad n \quad = \quad \frac{16}{30}$$

$$\frac{14}{30} \log 2 + \frac{16}{30}$$

Cross Entropy loss:

$$1 - \left(\frac{26}{30} \log_2 \frac{26}{30} + \frac{4}{30} \log_2 \frac{4}{30} \right) \quad \left. \vphantom{\frac{26}{30}} \right\} \text{entropy for 1st container}$$

=

$$1 - \left(\frac{14}{30} \log_2 \frac{14}{30} + \frac{16}{30} \log_2 \frac{16}{30} \right)$$

=

$$1 - \left(\frac{1}{30} \log_2 \frac{1}{30} + \frac{29}{30} \log_2 \frac{29}{30} \right)$$

Cross Entropy loss:

- it is also called logarithmic loss, log loss or logistic loss.
- Each predicted class probability is compared to the actual class desired output 0 or 1.
- A score/loss is calculated that penalizes the probability based on how far it is from the actual expected value.

$$L_{CE} = - \sum_{i=1}^n t_i \log(p_i).$$

t_i = truth label.
 p_i = softmax probability.
for i th class.

Binary Cross Entropy loss.

$$L = - \sum_{i=1}^2 t_i \log(P_i).$$

$$= - [t \log(p) + (1-t) \log(1-p)].$$

$\log_2 \rightarrow$ bits.

$\log_e \rightarrow$ nats.

$\log_2$ is faster to compute than the ~~de~~ logarithm. for cross-entropy compute per iter cost is 1, one for concern of fast.

Categorical Cross-Entropy.

Sparse Categorical Cross-Entropy.

$\rightarrow$ same loss function. but difference between the two. but de is on how 'truth' labels are defined.

$\rightarrow$ it use one-hot encoded

$[1\ 0\ 0]$

$[0\ 1\ 0]$

$[0\ 0\ 1]$

$\rightarrow$ in one integer encoded,

$[1]$

$[2]$

$[3]$